Algebraic Two-Electron Integrals on the Prolate Spheroidal Lattice*

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Paper 11 established the prolate spheroidal lattice as the natural geometry for diatomic molecules, solving H_2^+ to 0.0002% energy error (spectral Laguerre solver) with zero free parameters. The extension to two electrons requires electron-electron repulsion integrals $\langle\phi_i|1/r_{12}|\phi_j\rangle$ over the six-dimensional two-particle configuration space. The standard approach—numerical quadrature on a grid in prolate spheroidal coordinates—converges slowly due to the Coulomb singularity at $r_{12} = 0$ and saturates at $\sim$80% of the exact H_2 dissociation energy D_e regardless of basis size.

We replace numerical integration with algebraic evaluation using the Neumann expansion of $1/r_{12}$ in prolate spheroidal coordinates, which decomposes each six-dimensional integral into a finite sum of products of one-dimensional auxiliary integrals computable by recurrence relations. The resulting two-electron integrals are exact within the Neumann truncation order L, with no six-dimensional quadrature and no fitted parameters; the sole residual numerical integration is a single one-dimensional adaptive quadrature for the log-singular second-kind Legendre moment B_l.

A systematic convergence study on H_2 at $R = 1.4011$ bohr demonstrates that the Neumann algebraic V_{ee} yields 92.4% of the exact D_e with 27 basis functions, compared to 79.6% for numerical V_{ee} at the same basis size—a uniform 12–20 percentage point improvement across all basis sizes from 1 to 72 functions. The Neumann result plateaus at 92.4%, confirming that V_{ee} integration error has been eliminated. The remaining 7.6% gap is diagnosed as a one-electron basis completeness limit: the electron-electron cusp requires non-analytic terms ($r_{12}^{1/2}$, $r_{12}\ln r_{12}$) that no polynomial prolate spheroidal basis can represent. This diagnosis identifies the next natural geometry in the GeoVac program: hyperspherical coordinates for the three-body coalescence.

I. INTRODUCTION

The Geometric Vacuum program identifies and discretizes the *natural geometry* of each quantum system—the coordinate system where the Schrödinger equation separates—replacing continuous integration with algebraic operations on quantum number labels [1–3]. For atoms, the natural geometry is Fock's three-sphere S^3 [5], where the Coulomb potential disappears into the conformal metric and the Schrödinger equation becomes a pure Laplace–Beltrami eigenvalue problem. For diatomic molecules, the natural geometry is the prolate spheroid, where the two-center Coulomb problem separates into coupled ordinary differential equations [4].

Paper 11 implemented this principle for the one-electron molecule H_2^+, achieving 0.0002% energy error (spectral Laguerre solver) with zero free parameters. The extension to two electrons introduces a qualitatively new challenge: the electron-electron repulsion integral

$$V_{ee}^{ij} = \left\langle \phi_i \left| \frac{1}{r_{12}} \right| \phi_j \right\rangle, \tag{1}$$

which couples the coordinates of both electrons and cannot be evaluated by separation of variables. In the prolate spheroidal Hylleraas-type CI framework, these integrals are six-dimensional (three coordinates per electron, with azimuthal symmetry reducing the effective dimensionality).

The standard approach computes Eq. (1) numerically, using the complete elliptic integral $K(k)$ to perform the azimuthal average of $1/r_{12}$ and Gauss–Legendre quadrature for the remaining four dimensions $(\xi_1, \eta_1, \xi_2, \eta_2)$. This approach has two fundamental limitations:

1. The Coulomb singularity at $r_{12} = 0$ (the electron-electron coalescence point) is poorly resolved by polynomial quadrature, requiring increasingly fine grids for convergence.

2. The exponential decay $e^{-\alpha(\xi_1+\xi_2)}$ of the basis functions extends to $\xi \to \infty$, requiring truncation of the semi-infinite radial domain.

As we demonstrate in Section V, these limitations cause the numerical V_{ee} to saturate at $\sim$80% of the exact H_2 dissociation energy, regardless of how many basis functions are employed.

This paper replaces numerical quadrature with the Neumann expansion of $1/r_{12}$ in prolate spheroidal coordinates [6–8], which converts each V_{ee} matrix element into a finite sum of products of one-dimensional integrals. These auxiliary integrals—angular moments $C_l(p)$, exponential moments $A_n(\alpha)$, and first- and second-kind Legendre moments $A_l(p, \alpha)$ and $B_l(p, \alpha)$—are computed by recurrence relations and closed forms, with the sole exception of the log-singular second-kind moment B_l, which uses a single one-dimensional adaptive quadrature (Sec. III.D). The six-dimensional repulsion integral

* Paper 12 in the Geometric Vacuum series

is thereby reduced to a finite sum of algebraic operations on quantum number labels plus this one 1D quadrature.

The structure of this paper is as follows. Section II derives the Neumann expansion and its factorization for σ-state matrix elements. Section III defines the auxiliary integrals and their recurrence relations. Section IV describes the computational pipeline. Section V presents the convergence study comparing numerical and algebraic V_{ee}. Section VI diagnoses the origin of the remaining 7.6% gap. Section VII discusses implications for the GeoVac program. Section VIII concludes.

II. THE NEUMANN EXPANSION

A. Expansion of $1/r_{12}$ in prolate spheroidal coordinates

The Coulomb kernel $1/|\mathbf{r}_1-\mathbf{r}_2|$ can be expanded in prolate spheroidal coordinates (ξ, η, φ) using the Neumann expansion [6, 9]:

$$\frac{1}{r_{12}} = \frac{2}{R}\sum_{l=0}^{\infty}\sum_{m=0}^{l}(2-\delta_{m0})\frac{(l-m)!}{(l+m)!}P_l^m(\xi_<)\,Q_l^m(\xi_>)\,P_l^m(\eta_1)\,P_l^m(\eta_2)\,\cos m(\varphi_1-\varphi_2), \tag{2}$$

where $\xi_< = \min(\xi_1, \xi_2)$, $\xi_> = \max(\xi_1, \xi_2)$, P_l^m and Q_l^m are associated Legendre functions of the first and second kind, and R is the internuclear distance.

For ${}^1\Sigma_g^+$ states of homonuclear diatomics, the wavefunction has $m = 0$ symmetry. The azimuthal integration $\int_0^{2\pi} d\varphi_1 \int_0^{2\pi} d\varphi_2$ projects onto the $m = 0$ component:

$$\left\langle \frac{1}{r_{12}} \right\rangle_{\varphi} = \frac{8\pi^2}{R}\sum_{l=0}^{\infty}(2l+1)\,P_l(\xi_<)\,Q_l(\xi_>)\,P_l(\eta_1)\,P_l(\eta_2), \tag{3}$$

where $P_l \equiv P_l^0$ and $Q_l \equiv Q_l^0$ are ordinary Legendre functions. The factor $8\pi^2 = 2 \times 4\pi^2$ combines the $2/R$ prefactor of the Neumann expansion (2) with the $4\pi^2$ from $\int_0^{2\pi}\int_0^{2\pi}\cos 0\, d\varphi_1\, d\varphi_2$. The $(2l + 1)$ factor arises from the Legendre polynomial normalization in the angular completeness relation $\int_{-1}^{1} P_l(\eta)\, P_{l'}(\eta)\, d\eta = 2\delta_{ll'}/(2l + 1)$.

Equation (3) is the key identity: it separates the two-electron Coulomb kernel into a sum of products of functions of individual coordinates. Each term in the sum involves $P_l(\xi_<)Q_l(\xi_>)$, which couples the two radial coordinates, and $P_l(\eta_1)P_l(\eta_2)$, which factorizes completely.

B. Matrix element factorization

Consider two basis functions of the Hylleraas type in prolate spheroidal coordinates:

$$\phi_i(\xi_1, \eta_1, \xi_2, \eta_2) = \mathcal{N}_i\, e^{-\alpha(\xi_1+\xi_2)}\, \xi_1^{j_i}\xi_2^{k_i}\, \eta_1^{l_i}\eta_2^{m_i} + (1 \leftrightarrow 2), \tag{4}$$

where the $(1 \leftrightarrow 2)$ exchange ensures bosonic spatial symmetry for the singlet state, and (j_i, k_i, l_i, m_i) are non-negative integer quantum numbers. The exponential parameter α controls the spatial extent; for all basis functions in a given calculation we use a common α optimized variationally.

The V_{ee} matrix element between ϕ_i and ϕ_j is

$$V_{ee}^{ij} = \iint \phi_i^*\, \frac{1}{r_{12}}\, \phi_j\; J(\xi_1, \eta_1)\, J(\xi_2, \eta_2)\, d\xi_1\, d\eta_1\, d\xi_2\, d\eta_2, \tag{5}$$

where $J(\xi, \eta) = (R/2)^3(\xi^2 - \eta^2)$ is the prolate spheroidal Jacobian (with the 2π azimuthal factors already absorbed into Eq. (3)).

Substituting the Neumann expansion (3) and expanding the product of two basis functions, each unsymmetrized term contributes a product of the form

$$e^{-2\alpha(\xi_1+\xi_2)}\, \xi_1^{P_1}\xi_2^{P_2}\, \eta_1^{Q_1}\eta_2^{Q_2}, \tag{6}$$

where $P_1 = j_i + j_j$, $P_2 = k_i + k_j$, $Q_1 = l_i + l_j$, $Q_2 = m_i + m_j$ are the combined powers from bra and ket.

The Jacobian factor $(\xi^2 - \eta^2)$ for each electron expands the integral into four sub-terms with shifted powers:

$$\begin{aligned}
(+1) &: (P_1{+}2,\; P_2{+}2,\; Q_1,\; Q_2),\\
(-1) &: (P_1{+}2,\; P_2,\; Q_1,\; Q_2{+}2),\\
(-1) &: (P_1,\; P_2{+}2,\; Q_1{+}2,\; Q_2),\\
(+1) &: (P_1,\; P_2,\; Q_1{+}2,\; Q_2{+}2).
\end{aligned} \tag{7}$$

For each such sub-term with effective powers $(\tilde{P}_1, \tilde{P}_2, \tilde{Q}_1, \tilde{Q}_2)$, the matrix element factorizes as

$$\frac{\pi^2 R^5}{8}\sum_{l=0}^{L}(2l+1)\, X_l(\tilde{P}_1, \tilde{P}_2, 2\alpha)\, C_l(\tilde{Q}_1)\, C_l(\tilde{Q}_2), \tag{8}$$

where $C_l(q) = \int_{-1}^{+1} \eta^q\, P_l(\eta)\, d\eta$ are angular moment integrals, and

$$X_l(p_1, p_2, \alpha) = \int_1^{\infty}\int_1^{\infty} \xi_1^{p_1}\xi_2^{p_2}\, e^{-\alpha(\xi_1+\xi_2)}\, P_l(\xi_<)\, Q_l(\xi_>)\, d\xi_1\, d\xi_2 \tag{9}$$

is the two-dimensional radial integral coupling the two electrons through the $P_l(\xi_<)Q_l(\xi_>)$ kernel.

The prefactor $\pi^2R^5/8$ arises from $(2/R) \times (R/2)^6 \times (2\pi)^2 = \pi^2R^5/8$, combining the Neumann prefactor $2/R$, the two Jacobians $(R/2)^3$ each, and the azimuthal integration $4\pi^2$.

C. Selection rules

The angular integrals $C_l(q)$ obey a powerful selection rule:

$$C_l(q) = 0 \quad \text{if } q < l \text{ or } (q - l) \text{ is odd}. \tag{10}$$

This follows from the orthogonality of Legendre polynomials and parity: $P_l(\eta)$ has parity $(-1)^l$, so $\eta^q P_l(\eta)$

integrates to zero over $[-1,1]$ unless $q+l$ is even, i.e., $(q-l)$ is even. The condition $q \geq l$ ensures that η^q has enough powers to produce a nonzero overlap with P_l.

For a basis with maximum η-powers $q_{\max}$, the effective Neumann truncation is $l_{\mathrm{eff}} = \min(L, q_{\max})$, where L is the nominal truncation order. We use $L = 20$ throughout, which is more than sufficient since our basis functions have $q_{\max} \leq 8$.

III. AUXILIARY INTEGRALS AND RECURRENCE RELATIONS

The factorized form (8) reduces each V_{ee} matrix element to products of three families of one-dimensional integrals: $C_l(q)$, $A_n(\alpha)$ and its Legendre generalizations, and $B_l(p,\alpha)$. The C_l and A_n families are computed by pure recurrence relations; the log-singular second-kind moment B_l requires a single one-dimensional adaptive quadrature (below).

A. Angular moments $C_l(q)$

$$C_l(q) = \int_{-1}^{+1} \eta^q \, P_l(\eta) \, d\eta. \tag{11}$$

The base cases are

$$C_0(q) = \begin{cases} 2/(q+1) & q \text{ even}, \\ 0 & q \text{ odd}, \end{cases} \qquad C_1(q) = \begin{cases} 0 & q \text{ even}, \\ 2/(q+2) & q \text{ odd}. \end{cases} \tag{12}$$

Higher orders follow from the Legendre recurrence:

$$(l+1)\, C_{l+1}(q) = (2l+1)\, C_l(q+1) - l\, C_{l-1}(q). \tag{13}$$

The recurrence $C_{l+1}(q)$ requires $C_l(q+1)$, so computing $C_{l_{\max}}(q)$ requires workspace up to $q + l_{\max}$.

B. Exponential moments $A_n(\alpha)$

$$A_n(\alpha) = \int_1^\infty \xi^n \, e^{-\alpha\xi} \, d\xi. \tag{14}$$

The base case is $A_0(\alpha) = e^{-\alpha}/\alpha$, and the upward recurrence is

$$A_n(\alpha) = \frac{n\, A_{n-1}(\alpha) + e^{-\alpha}}{\alpha}. \tag{15}$$

For negative indices, $A_{-1}(\alpha) = E_1(\alpha)$ (the exponential integral) and the downward recurrence

$$A_{-n-1}(\alpha) = \frac{e^{-\alpha} - \alpha\, A_{-n}(\alpha)}{n} \tag{16}$$

extends to arbitrary negative n.

C. First-kind Legendre moments $A_l(p,\alpha)$

$$A_l(p,\alpha) = \int_1^\infty \xi^p \, e^{-\alpha\xi} \, P_l(\xi) \, d\xi. \tag{17}$$

The base cases are $A_0(p,\alpha) = A_p(\alpha)$ and $A_1(p,\alpha) = A_{p+1}(\alpha)$, since $P_0(\xi) = 1$ and $P_1(\xi) = \xi$. The Legendre recurrence gives

$$(l+1)\, A_{l+1}(p,\alpha) = (2l+1)\, A_l(p+1,\alpha) - l\, A_{l-1}(p,\alpha). \tag{18}$$

D. Second-kind Legendre moments $B_l(p,\alpha)$

$$B_l(p,\alpha) = \int_1^\infty \xi^p \, e^{-\alpha\xi} \, Q_l(\xi) \, d\xi. \tag{19}$$

These integrals involve the Legendre function of the second kind $Q_l(\xi)$, which has a logarithmic singularity at $\xi = 1$:

$$Q_0(\xi) = \tfrac{1}{2} \ln \frac{\xi+1}{\xi-1}, \qquad Q_l(\xi) \sim -\tfrac{1}{2} \ln(\xi-1) + O(1) \quad \text{as } \xi \to 1^+. \tag{20}$$

The log singularity is integrable (the integrand behaves as $\ln(\xi-1)\, e^{-\alpha\xi} \to \ln(\xi-1)\, e^{-\alpha}$ near $\xi = 1$), but it defeats polynomial quadrature rules such as Gauss–Laguerre, which assume smooth integrands.

We compute $B_l(p,\alpha)$ by adaptive numerical quadrature [16] with subdivision near $\xi = 1$ to resolve the log singularity. The cost is modest: each B_l evaluation takes $\sim$1 ms, and the full table $B_l(p,\alpha)$ for $l \leq 20$, $p \leq 12$, fixed α is computed in $\sim$0.2 s. Caching eliminates redundant computation across matrix elements at the same α.

In principle, the Legendre recurrence

$$(l+1)\, B_{l+1}(p,\alpha) = (2l+1)\, B_l(p+1,\alpha) - l\, B_{l-1}(p,\alpha) \tag{21}$$

could be used to generate B_l from B_0, but the log singularity affects *all* l values (not just $l = 0$), so the base case B_0 itself requires careful treatment. Adaptive quadrature is the most robust approach.

E. Two-dimensional radial integral $X_l(p_1,p_2,\alpha)$

The integral (9) couples the two radial coordinates through the $P_l(\xi_<) Q_l(\xi_>)$ kernel. We split it into two ordered regions:

$$X_l(p_1,p_2,\alpha) = \int_1^\infty \int_1^{\xi_2} \xi_1^{p_1} \xi_2^{p_2} \, e^{-\alpha(\xi_1+\xi_2)} P_l(\xi_1)\, Q_l(\xi_2) \, d\xi_1 \, d\xi_2 + (1 \leftrightarrow 2). \tag{22}$$

The inner integral $S_l(p,\alpha,\xi_0) = \int_1^{\xi_0} \xi^p\, e^{-\alpha\xi}\, P_l(\xi)\, d\xi$ is the *incomplete* first-kind Legendre moment. Expanding $\xi^p P_l(\xi) = \sum_j a_j\, \xi^j$ as a polynomial and integrating each monomial by parts against $e^{-\alpha\xi}$:

$$S_l(p,\alpha,\xi_0) = A_l(p,\alpha) - e^{-\alpha\xi_0} \sum_j a_j \sum_{k=0}^{j} \frac{j!}{(j-k)!\,\alpha^{k+1}}\, \xi_0^{j-k} + (\text{boundary at } \xi = 1). \tag{23}$$

Substituting into the outer integral and combining the $e^{-\alpha\xi_0}$ factor from S_l with the $e^{-\alpha\xi_2}$ factor from the outer integrand yields $e^{-2\alpha\xi_2}$ terms, which are evaluated as $B_l(\cdot, 2\alpha)$:

$$X_l = A_l(p_1)\, B_l(p_2) + A_l(p_2)\, B_l(p_1) - (\text{correction terms involving } B_l(\cdot, 2\alpha)). \tag{24}$$

The full expression involves sums over the polynomial coefficients of $\xi^{p_i} P_l(\xi)$ and is implemented as described in Section IV.

IV. IMPLEMENTATION

A. The algebraic V_{ee} pipeline

The computation of the V_{ee} matrix proceeds in four stages:

1. **Table precomputation.** For a given exponential parameter α and basis quantum numbers, compute:
   - The angular moment table $C_l(q)$ for $l \le l_{\text{eff}}$, $q \le 2q_{\max} + 2$, via the recurrence (13). Cost: $O(l_{\max} \cdot q_{\max})$, negligible.
   - The first-kind Legendre moments $A_l(p, 2\alpha)$ via the recurrence (18). Cost: $O(l_{\max} \cdot p_{\max})$.
   - The second-kind Legendre moments $B_l(p, 2\alpha)$ and $B_l(p, 4\alpha)$ via adaptive quadrature with caching. Cost: $O(l_{\max} \cdot p_{\max})$ quadrature calls, $\sim$0.2 s total.
2. X_l **table.** Compute $X_l(p_1, p_2, 2\alpha)$ for all $l \le l_{\text{eff}}$, $p_1, p_2 \le p_{\max} + l_{\max} + 2$, using Eq. (24) with the precomputed A_l and B_l tables. Cost: $O(l_{\max} \cdot p_{\max}^2 \cdot l_{\max})$ arithmetic operations.
3. **Matrix assembly.** For each pair (i, j) of basis functions, expand the bra$\times$ket product into unsymmetrized terms, apply the Jacobian expansion (7), and sum the Neumann series (8) using precomputed C_l and X_l tables. Cost per element: $O(l_{\max})$ (a single loop over l with table lookups).
4. **Symmetrization.** Exploit $V_{ee}^{ij} = V_{ee}^{ji}$ to compute only the upper triangle.

The total cost scales as $O(N_{\text{bf}}^2 \cdot l_{\max})$ for the matrix assembly, plus $O(l_{\max}^2 \cdot p_{\max}^2)$ for table precomputation. For $N_{\text{bf}} = 27$ and $l_{\max} = 8$ (the effective truncation), the V_{ee} matrix is computed in $\sim$1 s.

B. Hybrid Hamiltonian: Numba $T + V_{ne}$ plus algebraic V_{ee}

The one-electron integrals (kinetic energy T and nuclear attraction V_{ne}) are computed analytically using integration by parts for the kinetic energy derivatives and direct quadrature for V_{ne}, accelerated by Numba JIT compilation [15]. A dedicated Numba kernel `hamiltonian_tvne_p0_kernel` computes $T + V_{ne}$ without V_{ee}, avoiding the expensive elliptic integral computation that dominates the full numerical kernel.

The complete Hamiltonian is assembled as

$$H = H_{T+V_{ne}}^{(\text{Numba})} + V_{ee}^{(\text{Neumann})}, \tag{25}$$

where the first term uses grid-based quadrature (for the non-singular T and V_{ne} integrals) and the second uses the algebraic Neumann expansion (for the singular V_{ee} integrals). This hybrid approach plays to the strengths of each method: grids handle smooth integrands efficiently, while the Neumann expansion handles the Coulomb singularity exactly.

C. Computational cost comparison

For the V_{ee} matrix alone:

- **Numerical** (elliptic integral quadrature): $O(N_{\text{bf}}^2 \cdot N_\xi^2 \cdot N_\eta^2)$. At $N_\xi = 30$, $N_\eta = 20$: $\sim$3 s for $N_{\text{bf}} = 6$, scaling as $N_\xi^2 N_\eta^2$ per element.
- **Algebraic** (Neumann expansion): $O(N_{\text{bf}}^2 \cdot l_{\max} + l_{\max}^2 \cdot p_{\max}^2)$. The B_l quadrature is a one-time cost ($\sim$0.2 s), after which the matrix assembly is pure arithmetic.

The algebraic approach is not always faster in wall time (the B_l adaptive quadrature has non-trivial overhead), but it is *exact within the Neumann truncation*, whereas the numerical approach always carries grid truncation error.

V. CONVERGENCE STUDY

A. Computational setup

We compute the H_2 ground-state energy at $R = 1.4011$ bohr (near-equilibrium) using a Hylleraas-type CI with basis functions (4). The basis is parameterized by $(j_{\max}, l_{\max})$, which control the maximum powers of ξ and η respectively. The exponential parameter α is optimized variationally at each basis size using golden-section search over $\alpha \in [0.5, 2.5]$.

For each basis size, we compute the ground-state energy using both numerical and Neumann V_{ee}:

- **Numerical:** Full grid quadrature with elliptic integral $K(k)$ on a 30×20 grid ($N_\xi \times N_\eta$).

- **Neumann:** Algebraic V_{ee} via Eq. (8) with $L = 20$.

The exact non-relativistic H_2 energy at this geometry is $E_{\text{exact}} = -1.17475$ Ha [11], giving a dissociation energy $D_e = 0.17475$ Ha relative to two separated hydrogen atoms ($E = -1.0$ Ha).

B. Results

Table I presents the complete convergence study. The basis configurations range from a single function ($j = 0$, $l = 0$, $N = 1$) to 72 functions ($j = 3$, $l = 3$).

Three features of Table I deserve emphasis:

a. 1. Systematic improvement. The Neumann V_{ee} yields a lower (better) energy than numerical V_{ee} at *every* basis size, by 12–20 percentage points. This is not a sporadic improvement from fortuitous cancellation; it is a uniform advantage arising from the elimination of grid truncation error in V_{ee}.

b. 2. Numerical plateau at ∼80%. The numerical V_{ee} saturates at ∼80% D_e from $N = 27$ onward: adding basis functions from 27 to 72 improves the energy by only 0.5 percentage points (79.6% → 80.1%). This saturation is caused by grid truncation error in the V_{ee} integrals, which introduces a systematic positive bias in the electron-electron repulsion. The basis is flexible enough to represent a better wavefunction, but the inaccurate V_{ee} prevents it from being realized variationally.

c. 3. Neumann plateau at 92.4%. The Neumann V_{ee} also plateaus, but at a much higher level: 92.2% at $N = 27$ and 92.4% at $N = 72$. Since the Neumann V_{ee} is exact (within the $L = 20$ truncation, which is converged by the selection rule), this plateau is *not* caused by integration error. It identifies a genuine basis completeness limit, analyzed in Section VI.

C. The sweet spot: 27 basis functions

The data reveal a clear efficiency optimum at $N = 27$ ($j = 2$, $l = 2$): this configuration achieves 92.2% D_e in 705 s, whereas tripling the basis to $N = 72$ gains only 0.2 percentage points at 7.5× the computational cost. The diminishing returns beyond $N = 27$ reflect the fact that additional σ-type basis functions with higher ξ and η powers cannot access the missing physics (Section VI).

D. Grid dependence of numerical V_{ee}

To confirm that the numerical plateau is grid-limited, we performed a grid convergence study for the 6-function basis ($j = 1$, $l = 0$) at fixed $\alpha = 1.0$:

With the same 6 basis functions, D_e% increases from 66% to 85.7% as the grid is refined from 12×8 to 60×40. The convergence is slow (∼1.3 percentage points per grid doubling) and the cost grows quadratically. In contrast, the Neumann result for this basis is grid-independent at 61.8% D_e (the basis itself limits accuracy at 6 functions; the V_{ee} angular series terminates exactly, with only the single B_l moment carried by a 1D quadrature).

VI. DIAGNOSIS OF THE REMAINING 7.6% GAP

A. What the plateau tells us

The Neumann plateau at 92.4% establishes that the V_{ee} integration is converged: the Neumann series at $L = 20$ is exact for our basis (the selection rule (10) caps the effective l well below 20). The 7.6% gap is therefore a property of the *basis itself*, not of the integral evaluation.

The basis functions (4) are products of integer powers of ξ and η times a common exponential $e^{-\alpha\xi}$. These are smooth, analytic functions of the single-electron coordinates. No matter how many such functions we include, they span only the polynomial×exponential subspace of L^2.

B. The electron-electron cusp

The exact two-electron wavefunction $\Psi(\mathbf{r}_1, \mathbf{r}_2)$ satisfies the Kato cusp condition [12] at the electron-electron coalescence point $r_{12} = 0$:

$$\left.\frac{\partial \bar{\Psi}}{\partial r_{12}}\right|_{r_{12}=0} = \frac{1}{2}\,\bar{\Psi}(r_{12} = 0), \tag{26}$$

where $\bar{\Psi}$ is the spherically averaged wavefunction. This cusp produces a discontinuity in the first derivative of Ψ with respect to r_{12} and, more fundamentally, introduces non-analytic dependence on r_{12}.

The Fock expansion [13, 14] of the exact wavefunction near the two-electron coalescence point has the form

$$\Psi = \sum_{n=0}^{\infty} \sum_{k=0}^{\lfloor n/2 \rfloor} R^{n/2}\,(\ln R)^k\, f_{nk}(\hat{\Omega}), \tag{27}$$

where $R = (r_1^2 + r_2^2)^{1/2}$ is the hyperradius and $\hat{\Omega}$ represents the hyperangular coordinates. The $R^{1/2}$ and $R \ln R$ terms are fundamentally non-analytic: no finite sum of polynomial×exponential functions in prolate spheroidal coordinates can represent them.

C. Why r_{12} basis functions help only partially

The traditional approach to the cusp problem is to include explicit r_{12} dependence in the basis, as in the classic James–Coolidge [10] and Kolos–Wolniewicz [11] wavefunctions. We tested this by extending the basis with r_{12}^p ($p = 1, 2$) factors. The results are instructive: with 18

TABLE I. Convergence of H_2 ground-state energy at $R = 1.4011$ bohr. N is the number of basis functions. $D_e\%$ is the fraction of the exact dissociation energy $D_e = 0.17475$ Ha recovered. $\Delta E = E_{\text{Neumann}} - E_{\text{numerical}}$.

Config	N	E_{num} (Ha)	$D_e\%_{\text{num}}$	t_{num} (s)	E_{Neu} (Ha)	$D_e\%_{\text{Neu}}$	t_{Neu} (s)	ΔE (Ha)
$j=0$, $l=0$	1	−1.073192	41.9	1	−1.107907	61.8	3	−0.0347
$j=1$, $l=0$	3	−1.083438	47.8	8	−1.115282	66.1	9	−0.0318
$j=1$, $l=1$	6	−1.103485	59.3	32	−1.129606	74.3	32	−0.0261
$j=2$, $l=1$	12	−1.107960	61.9	152	−1.132373	75.9	131	−0.0244
$j=2$, $l=2$	27	−1.138836	79.6	767	−1.160961	92.2	705	−0.0221
$j=3$, $l=2$	46	−1.139329	79.8	2248	−1.161162	92.4	1907	−0.0218
$j=3$, $l=3$	72	−1.139792	80.1	6076	−1.161304	92.4	5271	−0.0215

TABLE II. Grid convergence of numerical V_{ee} for the 6-function basis at $R = 1.4011$ bohr, $\alpha = 1.0$.

Grid ($N_\xi \times N_\eta \times N_\phi$)	$D_e\%$	t (s)
$12 \times 8 \times 8$	66.0	1
$20 \times 14 \times 16$	76.2	6
$30 \times 20 \times 24$	81.4	45
$40 \times 26 \times 16$	83.7	83
$60 \times 40 \times 16$	85.7	447

functions including r_{12}^2 and angular terms ($l \leq 1$), the numerical V_{ee} reaches 86.8% D_e at the $20\times14\times16$ grid—but this *also* saturates with grid refinement. The r_{12} factors improve the wavefunction's ability to represent the cusp, but the V_{ee} integrals involving r_{12} are even harder to evaluate numerically (they become seven-dimensional integrals with an additional radial singularity).

The Neumann algebraic approach at 92.2% with $p = 0$ basis functions outperforms the r_{12} + numerical approach at 86.8% with $p \leq 2$ basis functions. This confirms the central thesis: *the cusp does not need r_{12}^p basis functions—it needs exact V_{ee} computed from quantum number algebra.* The r_{12} basis functions are a workaround for poor V_{ee} integration, not a solution to the underlying representation problem.

D. The missing geometry

The 7.6% gap identifies a *geometric* limitation. The prolate spheroidal coordinates $(\xi_1, \eta_1, \xi_2, \eta_2)$ are natural for the *one-electron* nuclear attraction problem (they separate the two-center Coulomb potential), but they are not natural for the *two-electron* coalescence (they do not separate $1/r_{12}$). The cusp is a property of the *inter-electron* coordinate r_{12}, which in prolate spheroidals is a complicated function of all four variables:

$$r_{12}^2 = \frac{R^2}{4}\left[(\xi_1^2-1)(1-\eta_1^2)+(\xi_2^2-1)(1-\eta_2^2)-2\sqrt{(\xi_1^2-1)(1-\eta_1^2)(\xi_2^2-1)(1-\eta_2^2)}+(\xi_1\eta_1-\xi_2\eta_2)^2\right]. \tag{28}$$

The cusp $\partial\Psi/\partial r_{12} \neq 0$ at $r_{12} = 0$ cannot be represented by smooth functions of $\xi_1, \eta_1, \xi_2, \eta_2$ individually. It requires either explicit r_{12} dependence (the traditional approach, but hard to integrate) or a coordinate system where the coalescence point has a simple representation.

The natural geometry for the three-body coalescence problem is the *hyperspherical* coordinate system $(\rho, \theta, \hat{\Omega})$, where $\rho = (r_1^2 + r_2^2)^{1/2}$ is the hyperradius and the cusp condition becomes a boundary condition in θ at $\theta = \pi/4$ (the $r_1 = r_2$ manifold). In this coordinate system, the Fock expansion (27) in $\rho^{1/2}$ and $\rho \ln \rho$ becomes a local expansion in the natural variable, amenable to lattice discretization.

VII. IMPLICATIONS FOR THE GEOVAC PROGRAM

A. The natural geometry hierarchy

Papers 0, 1, and 7 established S^3 as the natural geometry for one-center, one-electron systems. Paper 11 established prolate spheroids as the natural geometry for two-center, one-electron systems. This paper completes the analysis for two-center, two-electron systems and identifies the hierarchy of natural geometries:

1. **One-center, one-electron** (H, He^+): Three-sphere S^3 via Fock projection. The Schrödinger equation becomes a Laplace–Beltrami eigenvalue problem. Accuracy: $< 0.1\%$ [1, 2].

2. **Two-center, one-electron** (H_2^+): Prolate spheroidal coordinates (ξ, η, φ). The Schrödinger equation separates into two coupled ODEs. Accuracy: 0.0002% (spectral Laguerre) [4].

3. **Two-center, two-electron** (H_2): Prolate spheroidal $\times$ prolate spheroidal with Neumann-algebraic V_{ee}. One-electron physics is exact; electron correlation is resolved to 92.4%. *This paper.*

4. **Three-body coalescence**: Hyperspherical coordinates $(\rho, \theta, \hat{\Omega})$ for the interelectron cusp. *Identified but not yet implemented.*

Each level in this hierarchy addresses a qualitatively different physical singularity: the nuclear Coulomb singularity $(1/r)$, the two-center potential $(1/r_A + 1/r_B)$, the electron-electron Coulomb singularity ($1/r_{12}$), and the coalescence cusp (non-analytic r_{12} dependence). The GeoVac program identifies each singularity's natural coordinate system and discretizes the Laplace–Beltrami operator in that system.

B. Quantum chemistry without six-dimensional quadrature

The Neumann algebraic V_{ee} realizes a form of quantum chemistry in which the two-electron integrals are reduced to recurrence relations on quantum number labels, the sole residual numerical step being a single one-dimensional adaptive quadrature for the log-singular second-kind moment B_l. Combined with the analytical kinetic energy (integration by parts) and nuclear attraction (simple polynomial moments), the *entire* Hamiltonian matrix is constructed without any six-dimensional quadrature.

This has conceptual significance beyond computational efficiency: it demonstrates that the physics of electron-electron repulsion in prolate spheroidal coordinates is fully encoded in the algebraic structure of the Legendre functions P_l and Q_l, not in the spatial geometry of a quadrature grid. The Neumann expansion is not an approximation; it is the exact representation of $1/r_{12}$ in the prolate spheroidal basis, truncated only by the angular momentum selection rule imposed by the basis itself.

C. Comparison with the S^3 approach

The graph Laplacian FCI approach (Papers 0, 1, 7, and the FCI papers) constructs V_{ee} from Slater F^0 integrals on the S^3 density overlap [3], which are computed analytically from hydrogen radial wavefunctions. That approach also avoids numerical integration of V_{ee} and achieves comparable accuracy for atoms.

The prolate spheroidal approach with Neumann V_{ee} extends this philosophy to molecules: the natural coordinate system for each class of problems determines the algebraic structure of the two-electron integrals. On S^3, the algebraic structure is the Slater expansion in radial quantum numbers. On the prolate spheroid, it is the Neumann expansion in Legendre orders. Both are special cases of the general principle: *in the natural coordinate system, the Coulomb kernel has a known multipole expansion that converts integrals into algebraic sums.*

VIII. CONCLUSION

We have demonstrated that the Neumann expansion of $1/r_{12}$ in prolate spheroidal coordinates eliminates the electron-electron integration bottleneck in the two-electron prolate spheroidal CI for H_2. The key results are:

1. **92.4% of exact** D_e with algebraic V_{ee} (a single 1D quadrature for the log-singular B_l moment) and zero fitted parameters.

2. **12–20 percentage point improvement** over numerical V_{ee} at every basis size from 1 to 72 functions.

3. **Numerical** V_{ee} **saturates at ∼80%** regardless of basis size, confirming grid truncation error as the dominant error source.

4. **27 basis functions is the sweet spot**: 92.2% in 705 s versus 92.4% in 5271 s at 72 functions.

5. **The 7.6% gap** is precisely diagnosed: it is a basis completeness limit, not an integration error. The electron-electron cusp requires non-analytic terms ($r_{12}^{1/2}$, $r_{12}\ln r_{12}$) that no polynomial prolate spheroidal basis can represent. This identifies hyperspherical coordinates as the next natural geometry in the GeoVac hierarchy.

The result demonstrates a form of quantum chemistry where recurrence relations on quantum number labels replace the six-dimensional quadrature for electron-electron repulsion, leaving only a single one-dimensional quadrature for the log-singular B_l moment. Just as Paper 7 showed that atomic energies are encoded in the topology of S^3, this paper shows that molecular electron-electron repulsion is encoded in the algebraic structure of the Legendre functions P_l and Q_l on the prolate spheroid. The physics is in the algebra, not in the grid.

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Appendix A: Numerical Verification of Auxiliary Integrals

All auxiliary integrals were validated against independent numerical quadrature. The test suite (`tests/test_neumann_vee.py`, 32 tests) verifies:

- $C_l(q)$: Base cases (C_0 even/odd, C_1 odd), selection rule, and known analytical values. Agreement to machine precision ($< 10^{-14}$).

- $A_n(\alpha)$: Comparison with `scipy.integrate.quad`. Agreement to $< 10^{-12}$.

- $A_l(p, \alpha)$: Comparison with numerical quadrature of $\xi^p e^{-\alpha\xi} P_l(\xi)$. Agreement to $< 10^{-10}$.

- $B_l(p, \alpha)$: Comparison with independent adaptive quadrature. Agreement to $< 10^{-10}$.

- $X_l(p_1, p_2, \alpha)$: Comparison with 2D numerical quadrature. Agreement to $< 10^{-8}$. Symmetry $X_l(p_1, p_2) = X_l(p_2, p_1)$ verified to $< 10^{-14}$.

- V_{ee} **matrix**: Symmetry, positive diagonal, and comparison with numerical elliptic integral quadrature. Neumann vs. numerical agreement to $<$ 0.01 Ha per element (the discrepancy is the grid error in the numerical values).

- **Full Hamiltonian**: Variational principle (energy above exact), bound state, and consistency between Python and Numba code paths ($< 10^{-8}$ Ha).

Appendix B: Comparison with r_{12}-Dependent Basis Functions

We tested basis functions with explicit r_{12}^p dependence ($p = 1, 2$), following the James–Coolidge approach [10]. The r_{12} factor is expressed in prolate spheroidal coordinates as a function of $(\xi_1, \eta_1, \xi_2, \eta_2, \varphi_1 - \varphi_2)$, making the integrals five-dimensional after azimuthal averaging.

Table III compares the best r_{12}-dependent results with the $p = 0$ Neumann results.

TABLE III. Comparison of r_{12}-dependent basis (numerical V_{ee}) vs. $p = 0$ basis (Neumann V_{ee}). All results at R = 1.4011 bohr.

Method	N	D_e%	t (s)
r_{12}^1, num., 60×40 grid	6	85.7	447
r_{12}^{0-2}, num., 20×14 grid	18	86.8	970
$p=0$, Neumann, 30×20 grid	27	92.2	705
$p=0$, Neumann, 30×20 grid	72	92.4	5271

The Neumann $p = 0$ result at 92.2% exceeds the best r_{12}-dependent numerical result (86.8%) by 5.4 percentage points, confirming that exact V_{ee} evaluation is more important than r_{12}-dependent basis functions when the V_{ee} integrals are computed numerically. The combination of r_{12}-dependent basis functions with Neumann-type algebraic V_{ee} is a natural extension left for future work.

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